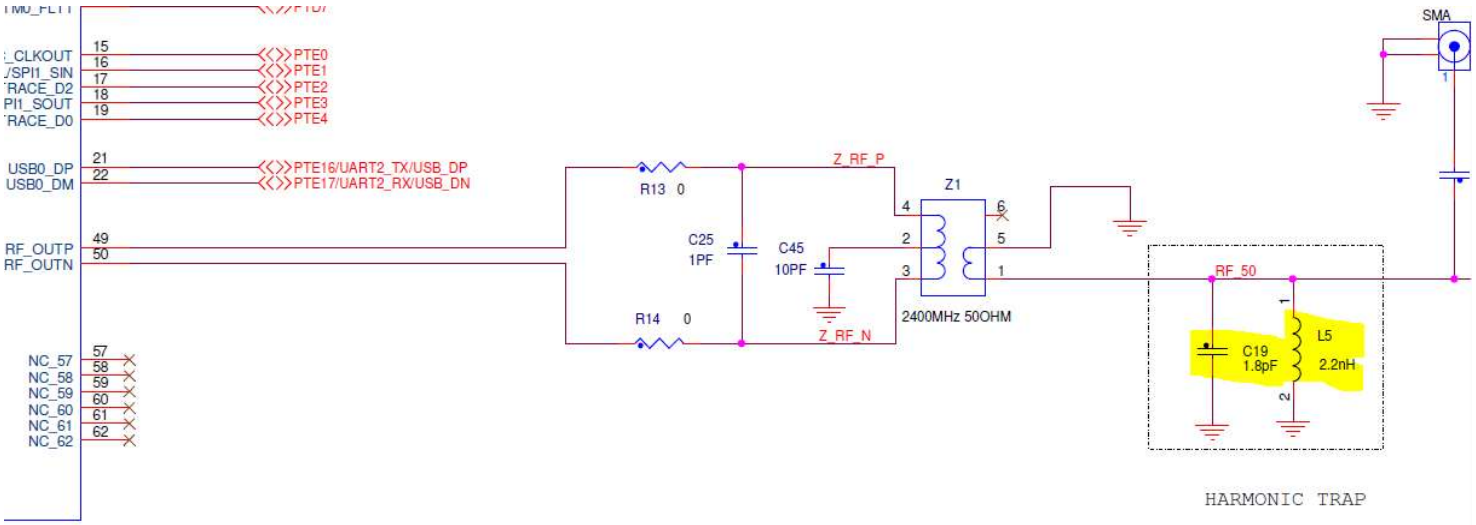
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Recalculate the parallel LC network originally used around 2.4 GHz to operate at 850 MHz, in order to find a commercial part number for C19 and L5, taking into account that the values must change according to the following requirements:



$F := 850\ \text{MHz}$ * *Transmition frequency*

$BW := 200\ \text{MHz}$ * *Bandwidth (-/+100Hz)*

$RF := 50\ \text{ohm}$ * *Transmition line impedance*

$Z1 := 50\ \text{ohm}$ * *Power supply impedance*

$P_{max} := 1\ \text{watt}$ * *Maximum power*

$V_{max} := 20\ \text{V}$ * Maximum transmission voltage (peak to peak)

$T := 25\ ^\circ\text{C}$ * Nominal operation temperature

* Resonance frequency

$$\omega_0 := 2 \cdot \pi \cdot F = 5.3407 \cdot 10^{9}\ \frac{\text{rad}}{\text{sec}}$$

* Q factor for given bandwith

$$Q := \frac{F}{BW} = 4.25$$

* Impedance

$$R_p := \frac{1}{\frac{1}{Z1} + \frac{1}{RF}} = 25\ \Omega$$

* Inductance and capacitance for given Q and RP

$$Q_L := \frac{R_p}{\omega_0 \cdot L} \quad \text{thus,} \quad L := \frac{R_p}{\omega_0 \cdot Q} = 1.1014\ \text{nH}$$

$$Q_C := \omega_0 \cdot C \cdot R_p \quad \text{thus,} \quad C := \frac{Q}{\omega_0 \cdot R_p} = 31.831\ \text{pF}$$

Position	Commercial value candidate	Part number	Stock observed
C19	33 pF, ±2%, C0G, 50 V	Murata **GJM1555C1H330GE01D**	146,705
L5	1.1 nH, ±0.1 nH	Abracon **ATFC-0402-1N1B-T**	63,533

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